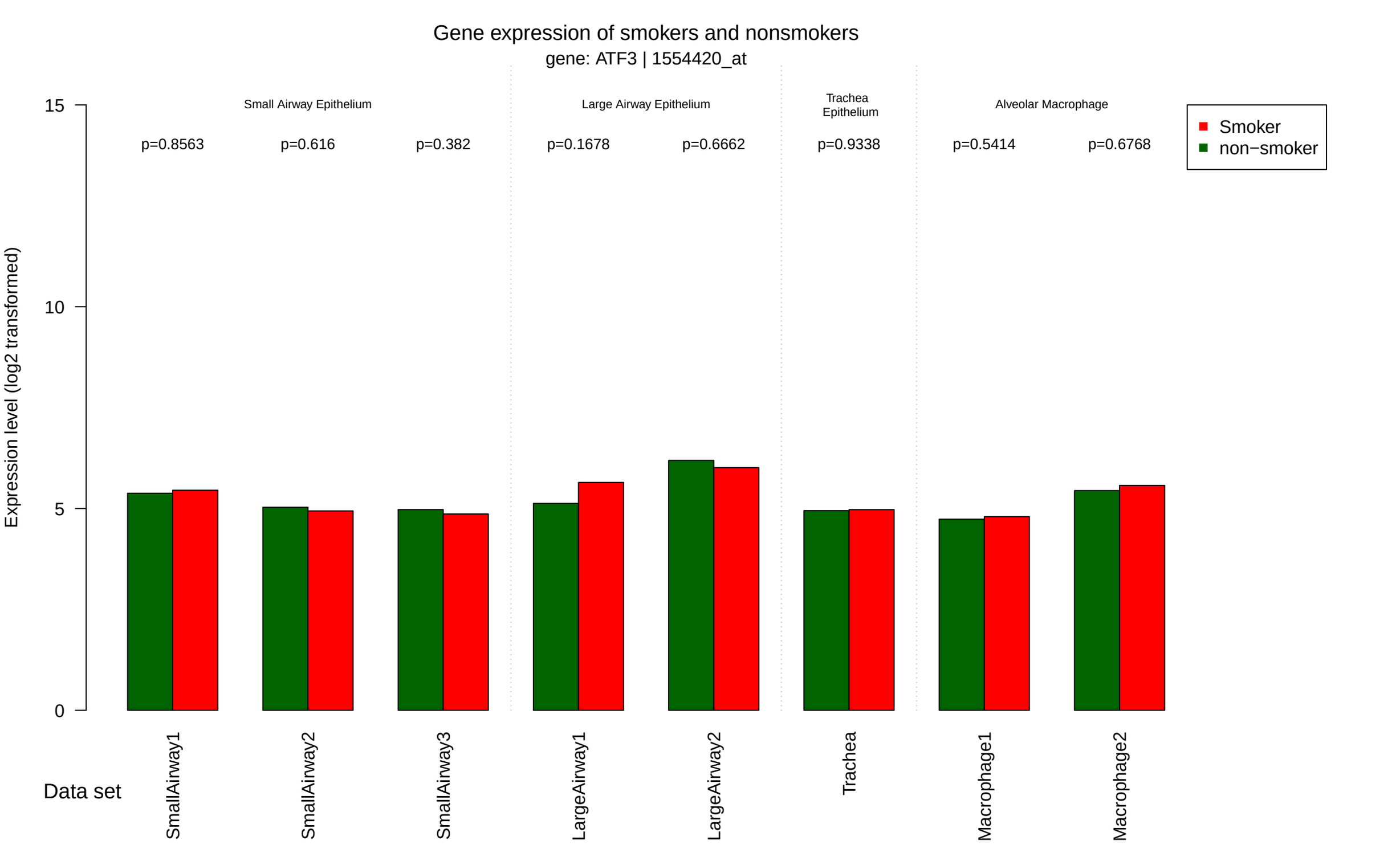
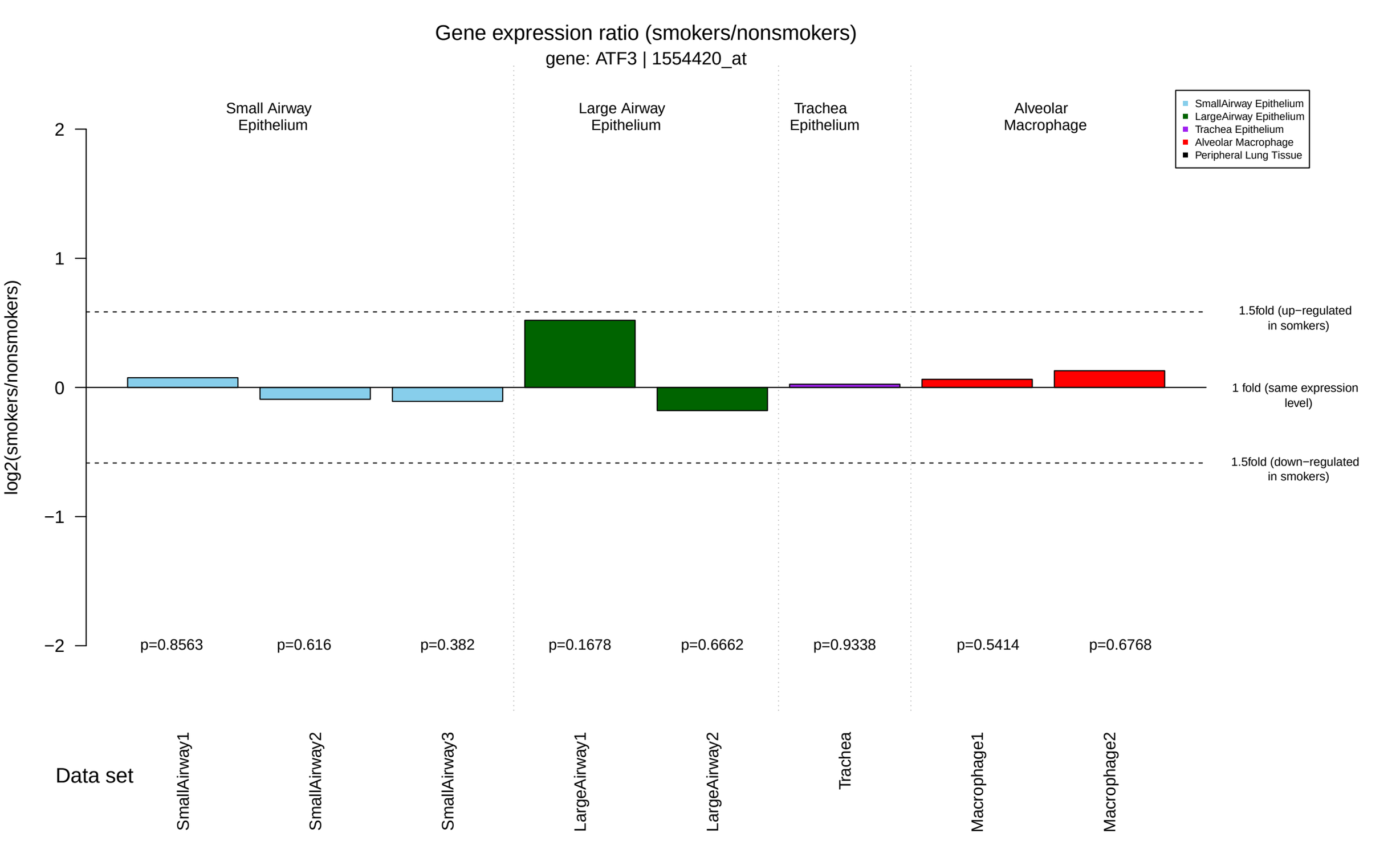
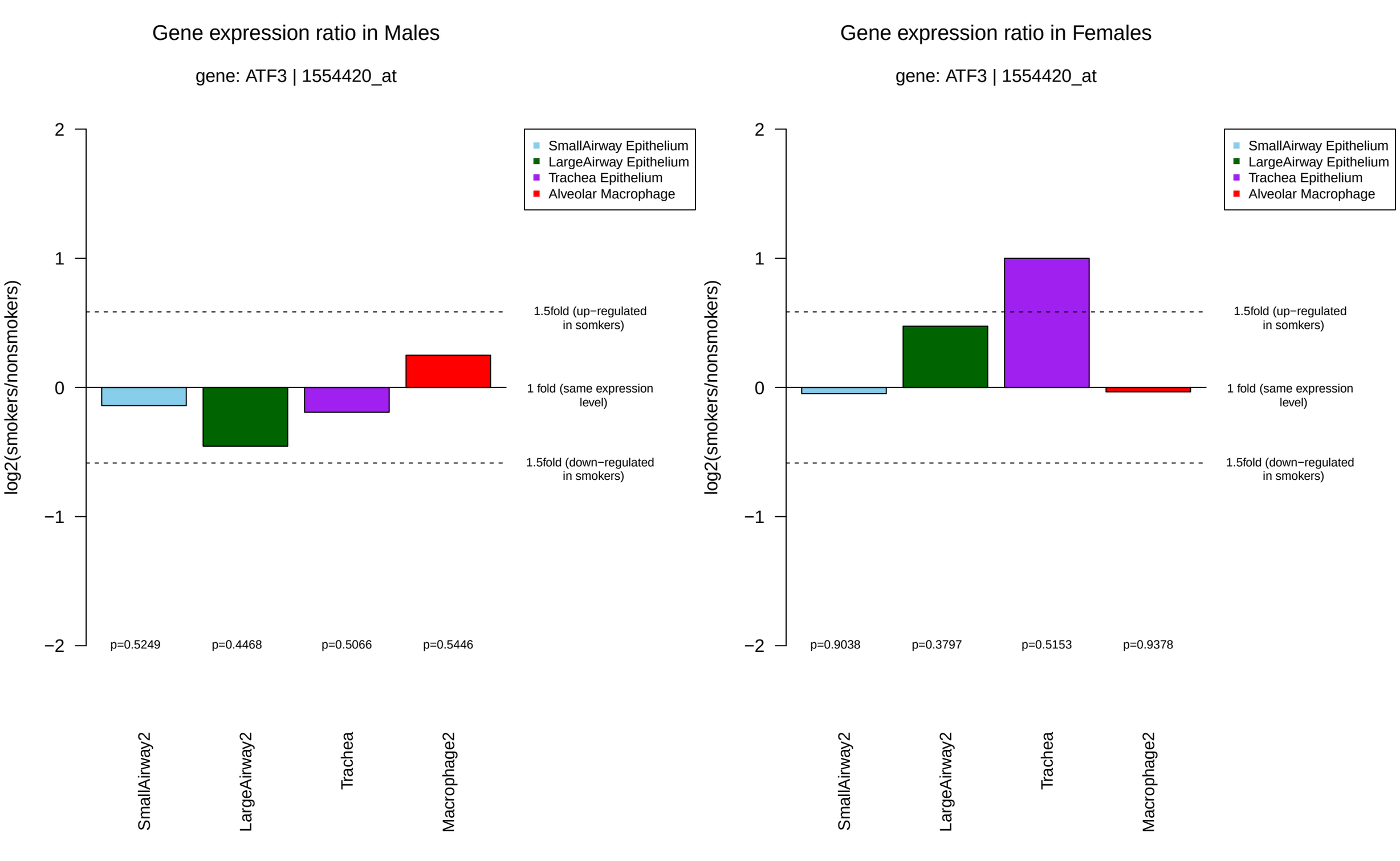
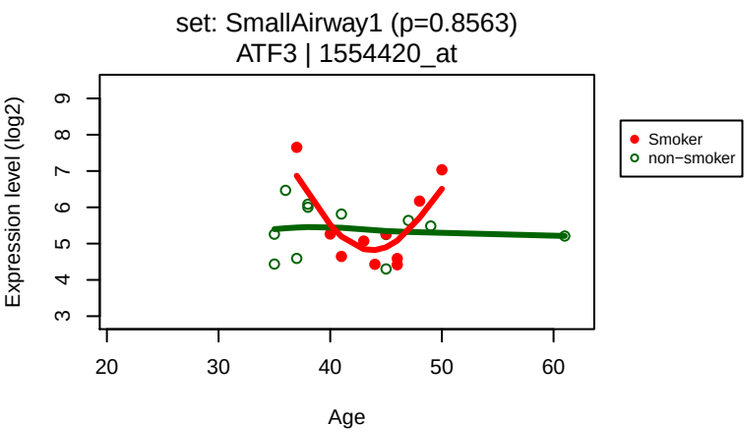
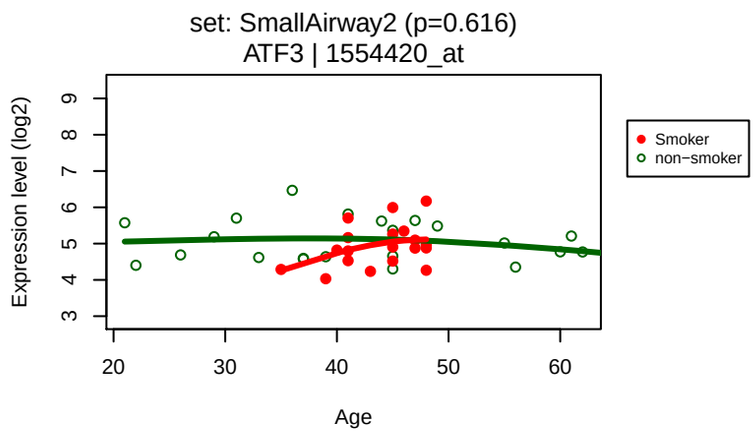
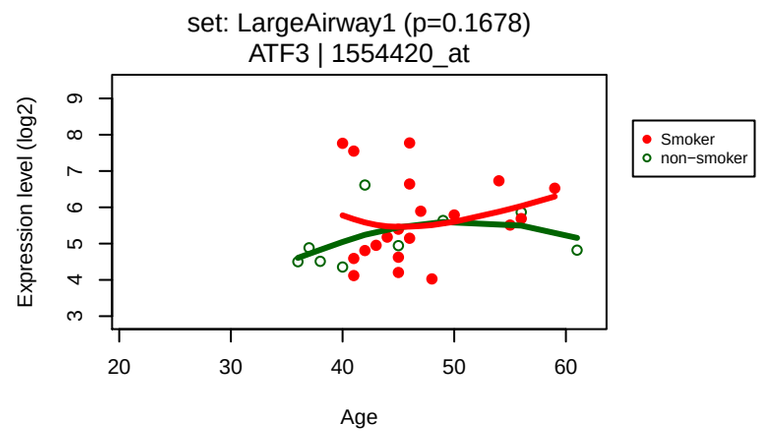

Gene expression of smokers and nonsmokers
gene: ATF3 | 1554420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8563
p=0.616
p=0.382
p=0.1678
p=0.6662
p=0.9338
p=0.5414
p=0.6768
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

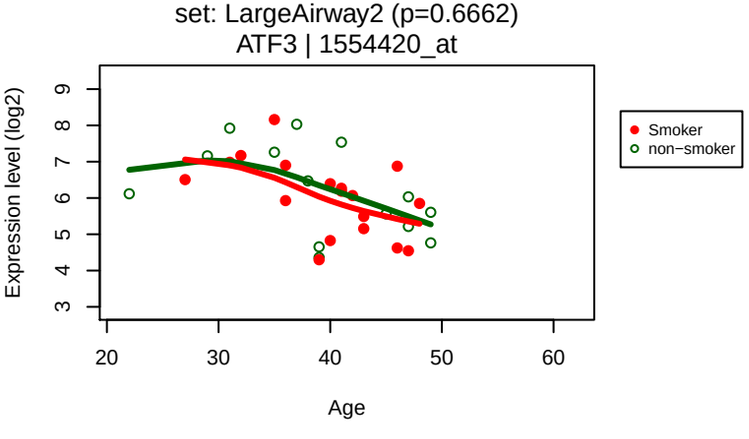

Gene expression ratio (smokers/nonsmokers)
gene: ATF3 | 1554420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8563
p=0.616
p=0.382
p=0.1678
p=0.6662
p=0.9338
p=0.5414
p=0.6768
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

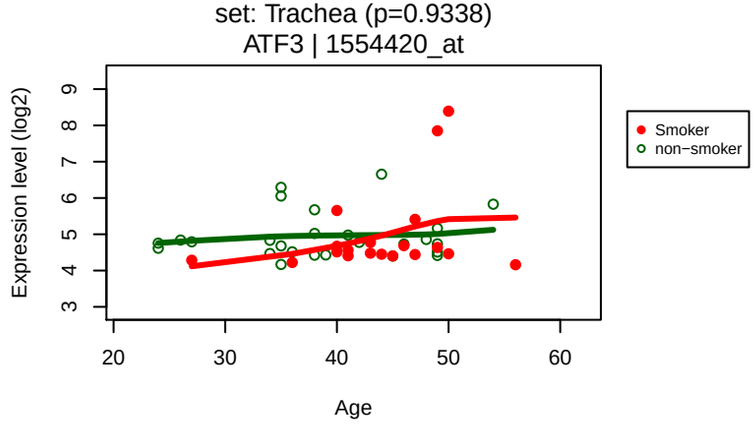

Gene expression ratio in Males
gene: ATF3 | 1554420_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5249
p=0.4468
p=0.5066
p=0.5446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF3 | 1554420_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9038
p=0.3797
p=0.5153
p=0.9378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

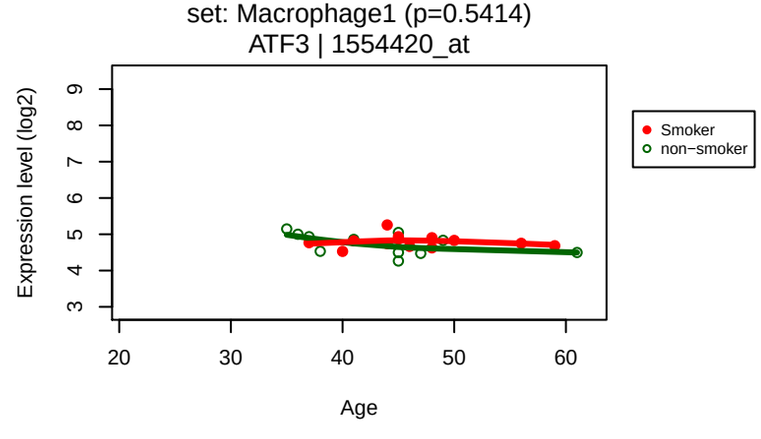

set: SmallAirway1 (p=0.8563)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker

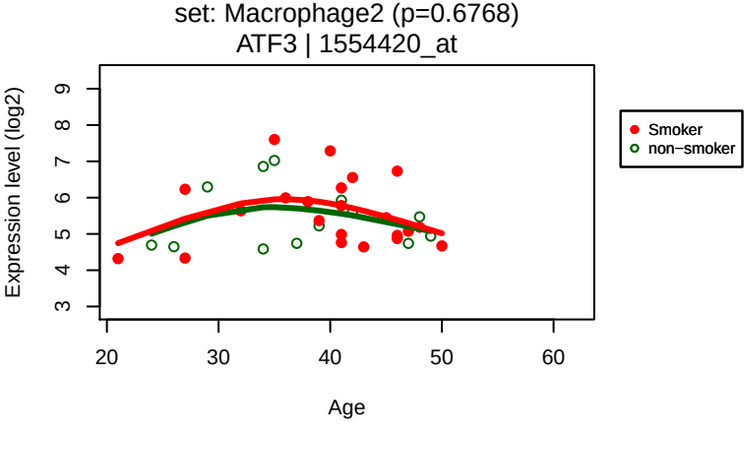

set: SmallAirway2 (p=0.616)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker

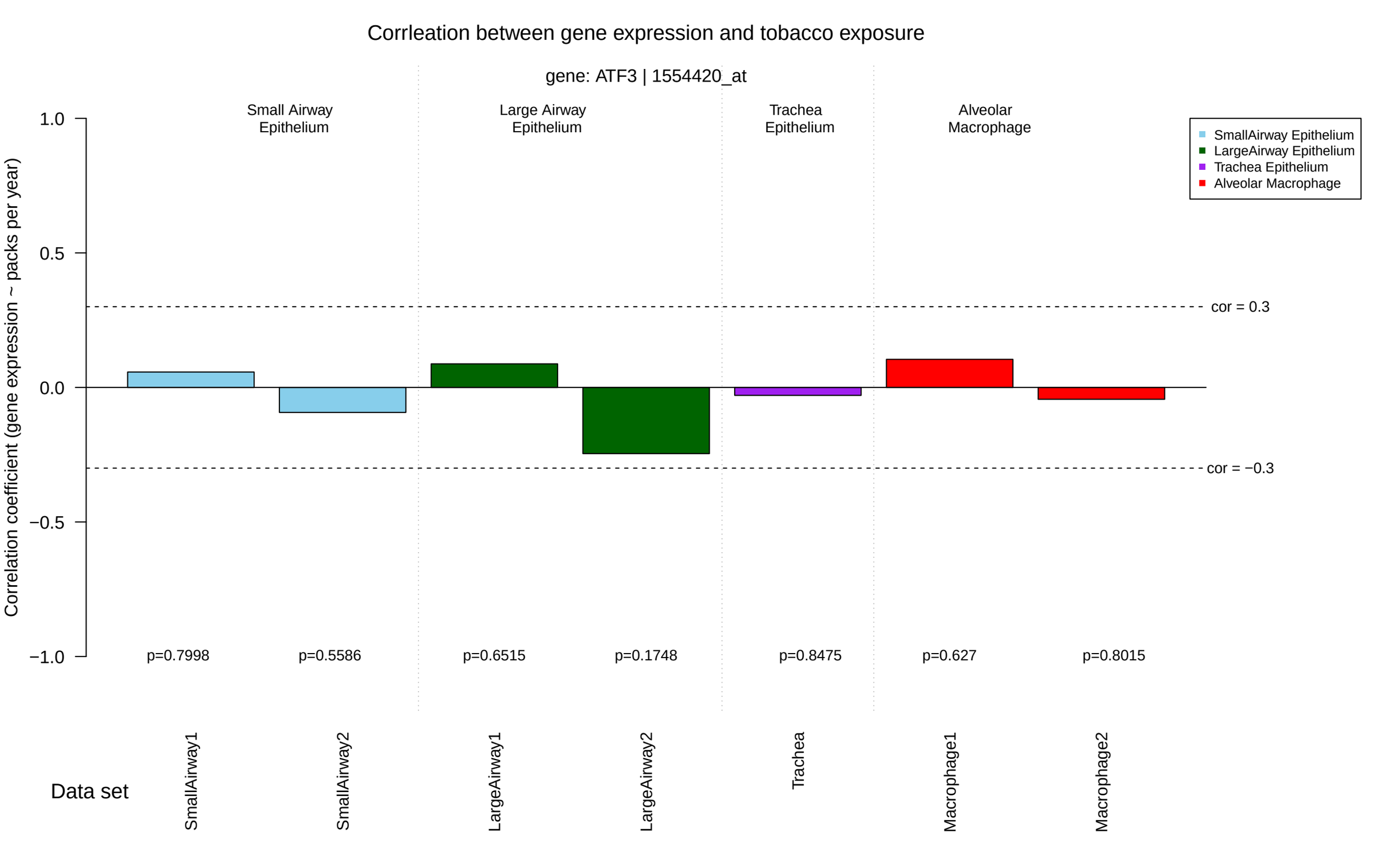

set: LargeAirway1 (p=0.1678)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6662)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9338)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5414)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6768)
ATF3 | 1554420_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF3 | 1554420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7998
p=0.5586
p=0.6515
p=0.1748
p=0.8475
p=0.627
p=0.8015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2